ELM: matched ion and Xe–He response
4 mm pitch, shared 5–145 cm placement

NAUSICAA

Quantity	Value
Matched events	8000
Corrected CSA maximum fast-model error (ENC)	0.43361
CSA maximum solver-refinement error (ENC)	0.00002
Mean initial pairs/event	112,163.29
Mean detected optical PE/event	507,460.27
Ion CSA total noiseless bias (%)	-0.67844
Ion CSA total signed reconstructed bias (%)	-0.69802
Events with a 50% compressed CSA history (%)	10.175
SiPM optical charge loss (%)	0.07616
Electron ASIC charge loss (%)	0.06607
Electron events with clipping (%)	5.100
Input-positive electron pulses clipped (%)	0.00379
Electron events retained	7592
Electron retained fraction (%)	94.900

Stage	Unit	Mean average	Mean median	Max average	Max median
Ion input	ions	392.723	334.626	7,014.820	6,447.000
Noiseless CSA	ion equivalents	388.885	333.378	6,820.894	6,356.719
Signed CSA output	ion equivalents	388.907	333.599	6,821.760	6,353.889
Blue photons at SiPM	photons	256.335	190.896	4,434.382	2,734.000
PDE optical triggers	PE	82.028	61.093	1,425.667	884.000
SiPM optical charge	PE equivalents	81.927	61.064	1,416.676	881.844
Signed electron ASIC	PE equivalents	81.915	61.253	1,335.749	882.619

Stage	All-event RMS (%)	Retained RMS (%)
Detected optical PE	0.19499	0.19612
After SiPM	0.20781	0.20174
Noisy, before clipping	0.29358	0.29171
After electron ASIC	0.56737	0.29171

Event ID	Electron drift (cm)	Maximum input PE	SiPM loss (%)	ASIC loss (%)
334	5.54	13,682	0.7052	10.6932
1277	6.61	9,206	0.5396	7.3965
7681	5.03	11,388	0.5445	7.3755
3444	6.49	11,690	0.4765	6.9584
2946	5.80	12,680	0.5795	6.7681
1689	8.27	10,122	0.5516	6.5477
6029	5.47	10,052	0.4992	6.3493
7731	5.25	11,766	0.4758	6.1554
2615	5.57	10,487	0.4876	6.1366
4264	6.01	8,809	0.4699	5.5297

Shared production

8,000 bb0nu events share one saved realization. Ion reference drift is uniform over 5–145 cm in a 150 cm chamber; electron reference drift is its exact complement. Primary-pair draws are anchored to the existing reference card, so changing placement does not change the pair realization. Event IDs, primary checksums and both distances are validated across common realization, ion collection, photon collection, corrected CSA, SiPM charge and electron ASIC output. The 4 mm ion pitch and 40 ms frames correspond to 4 mm drift buckets. Electron readout uses a 4 mm hole pitch and 250 ns (0.25 mm) buckets.

Electron optical and response model

Xe–15% He at 15 bar, E/P = 2 kV/cm/bar, 2.5 mm diameter holes/guides and 5 mm EL depth. A 0.5 mm radial xenon gap surrounds each guide. Unit electron collection and absorbing EL walls are assumed. Analytical hole-exit acceptance, TPB conversion 0.65, forward factor 0.5 and meridional capture $1-1/1.5$ give total optical probability 0.01187461. Guide propagation and NOA 87 coupling are ideal. Saved photons are blue photons at the SiPM; PDE is applied once afterward.

S14160-3015PS: 3×3 mm^2, 39,984 cells, 21,808 illuminated by the centered 2.5 mm disk, PDE 0.32, gain 360,000 and total dark rate 700 kcps. Illuminated and dark-only cell populations retain independent recovery states. The baseline uses a provisional 12.5 ns recovery constant with 128 statistical substeps per bucket. Electron electronics adds 3.5417 PE RMS and clips at 1 mA (4334.38 PE per bucket). Crosstalk, afterpulsing and gain spread remain deferred.

Ion electronics

The ion arm uses the calibrated CSA v3 circuit, 5 ms averaging and the 40 ms response/noise covariance. The linear fast path is validated against transistor controls at 0.5 ENC tolerance; histories below the 0.63 V guard receive direct LTspice treatment, with adaptive controls and solver refinement. The simulator binary and circuit calibration are reused only with matching content hashes. The signed three-frame inverse reconstructs ion charge, including quiet/recovery frames and temporal noise correlations. A 50% compression diagnostic identifies events containing such a pad history; it is separate from electron ASIC clipping.

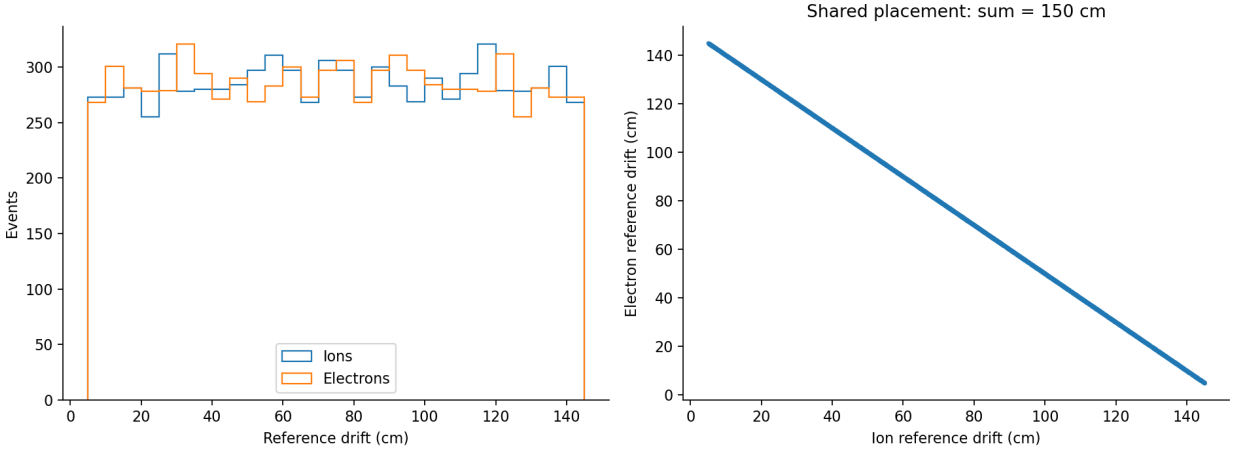
Definitions

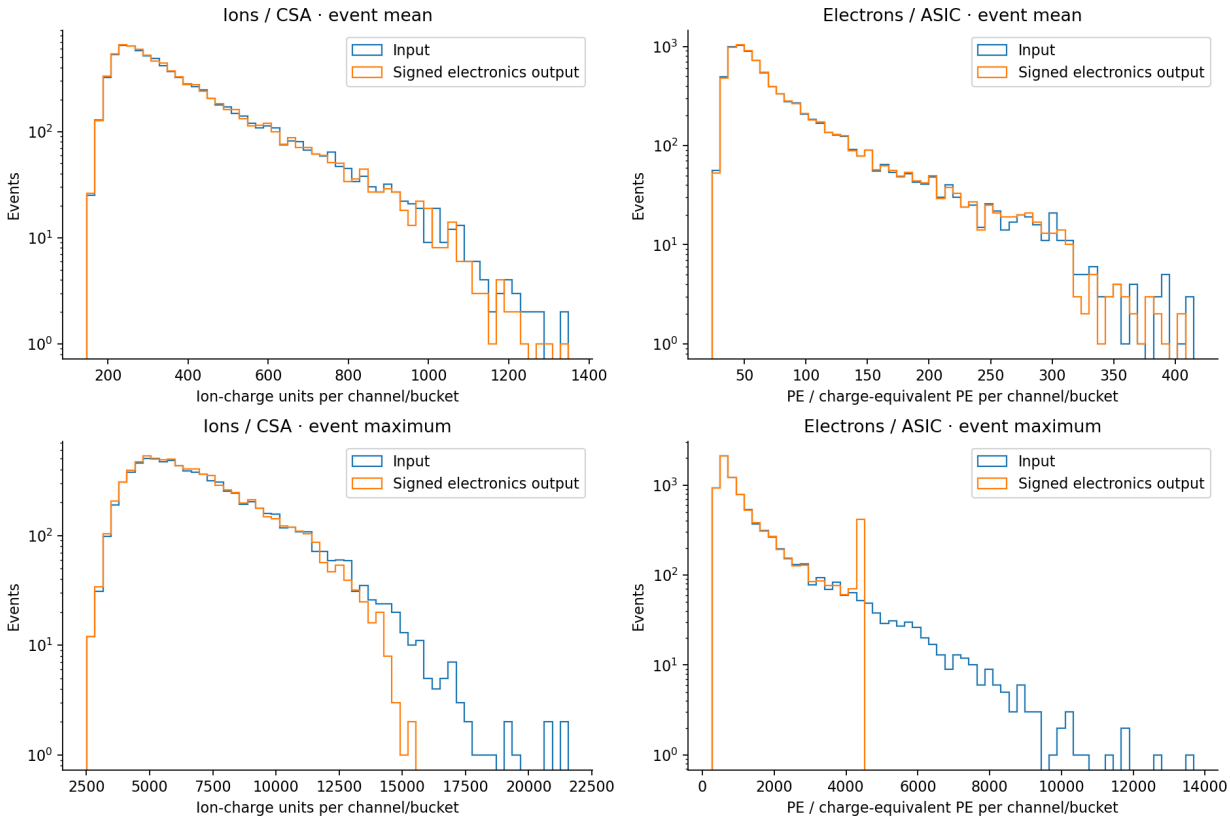
Pulse tables use all input-positive channel/bucket cells, preserving those same cells through electronics. Each event contributes one mean and maximum. Signed response values remain intact. Ion total charge bias sums complete stored histories relative to collected ions. Electron energy widths are

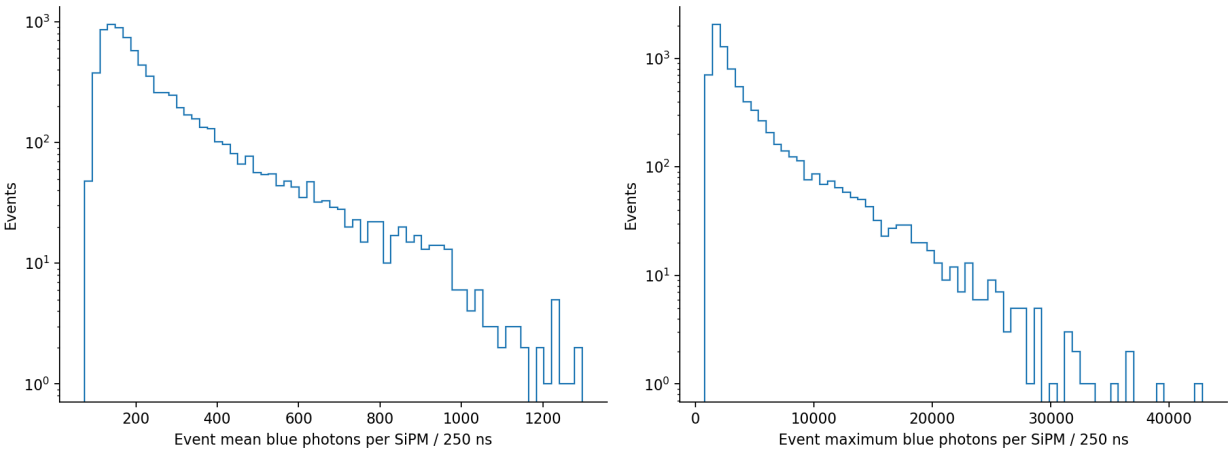
full-distribution RMS/mean after correcting known attachment survival and dividing by deposited MC energy; noisy totals subtract expected dark charge. The retained electron sample rejects an event when any electron ASIC bucket clips.

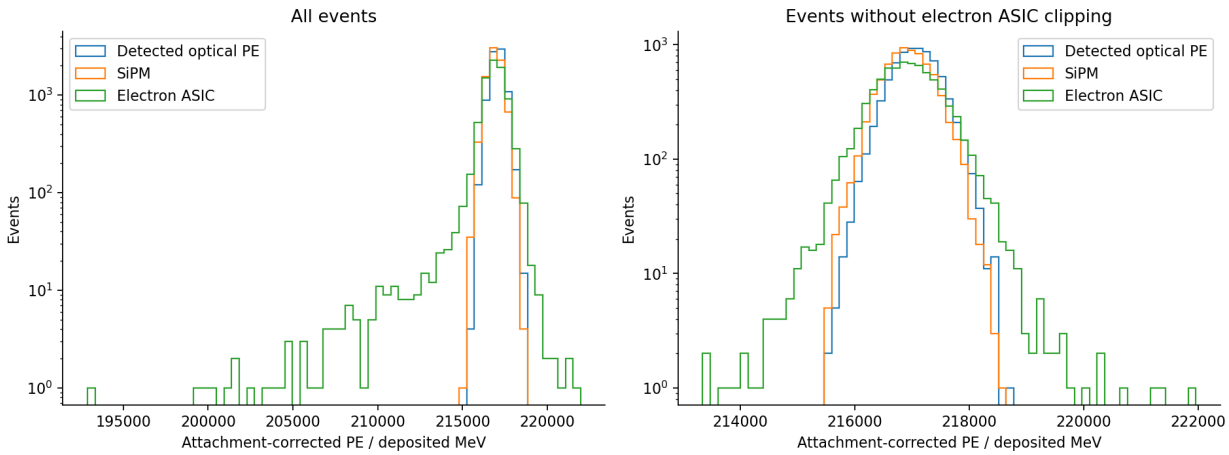
Timing

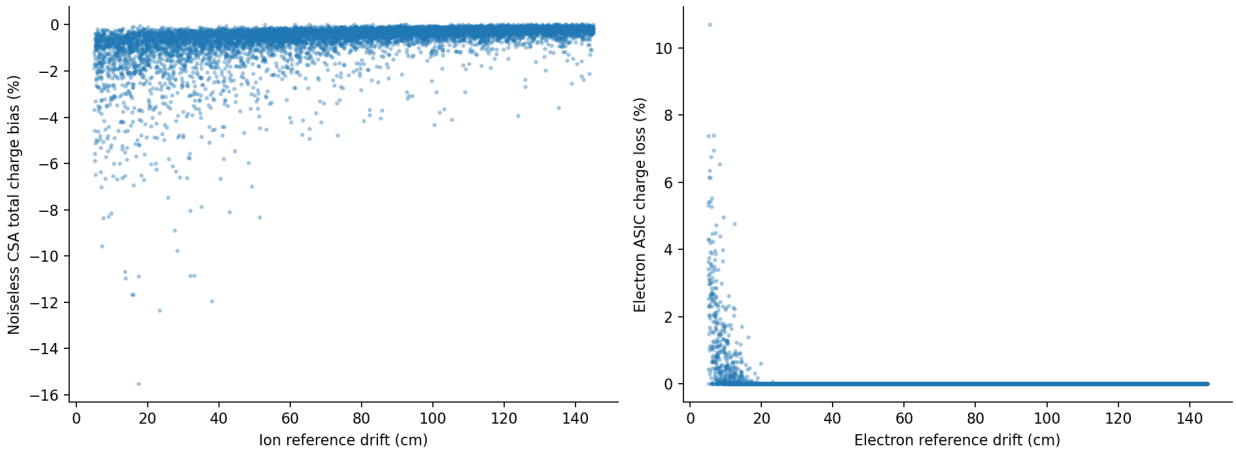
Detector production: 128.08 s; concurrent SiPM/ASIC: 484.26 s; concurrent CSA pipeline: 1733.83 s; complete paired production and validation: 1877.37 s. Reduction and presentation are separate.


Events
Reference drift (cm)
Ions
Electrons
Shared placement: sum = 150 cm
Electron reference drift (cm)
Ion reference drift (cm)


Ions / CSA · event mean
Input
Signed electronics output
Events
Ion-charge units per channel/bucket
Electrons / ASIC · event mean
Input
Signed electronics output
Events
PE / charge-equivalent PE per channel/bucket
Ions / CSA · event maximum
Input
Signed electronics output
Events
Ion-charge units per channel/bucket
Electrons / ASIC · event maximum
Input
Signed electronics output
Events
PE / charge-equivalent PE per channel/bucket


Events
10^3
10^2
10^1
10^0
200
400
600
800
1000
1200
Event mean blue photons per SiPM / 250 ns
Events
10^3
10^2
10^1
10^0
0
10000
20000
30000
40000
Event maximum blue photons per SiPM / 250 ns


All events
Detected optical PE
SiPM
Electron ASIC
Events
Attachment-corrected PE / deposited MeV
Events without electron ASIC clipping
Detected optical PE
SiPM
Electron ASIC
Events
Attachment-corrected PE / deposited MeV


Noiseless CSA total charge bias (%)
0
−2
−4
−6
−8
−10
−12
−14
−16
0
20
40
60
80
100
120
140
Ion reference drift (cm)
Electron ASIC charge loss (%)
10
8
6
4
2
0
0
20
40
60
80
100
120
140
Electron reference drift (cm)

Reproduction

`results/elm_matched_production/manifest.toml` resolves all physical histories and their matching validation. Numerical reductions are `matched_analysis.h5` and `electron_analysis.h5` in that directory. Full workflow: `make elm-matched-production`; presentation: `make elm-matched-report`.

Resolved cards

```
schema = "nausicaa.electron_detector_spec.v1"
pressure_bar = 15.0
helium_fraction = 0.15
drift_speed_mm_us = 1.0
lifetime_ms = 15.0
hole_pitch_mm = 4.0
hole_diameter_mm = 2.5
hole_depth_mm = 5.0
sipm_shape = "square"
sipm_side_mm = 3.0
bucket_width_us = 0.25
reduced_field_kV_cm_bar = 2.0
j_variance_ratio = 0.030
j_status = "user_specified"
j_reference = "Oliveira Figure 6; corrected approximate user reading, 2026-11-04"

optical_model = "elm_blue_meridional"
electron_collection_efficiency = 1.0

[optics]
photon_band = "blue_tpb"
tpb_conversion_efficiency = 0.65
forward_fraction = 0.5
guide_refractive_index = 1.5
gap_refractive_index = 1.0
guide_length_mm = 100.0
gas_gap_radial_mm = 0.5
propagation_transmission = 1.0
coupling_transmission = 1.0
electrode_transmission = 1.0
coupling_adhesive = "Norland NOA 87"
guide_length_policy = "fixed_grading_field"
grading_field_kV_cm = 4.65
cathode_potential_V = 1200.0
entry_potential_V = 31500.0
exit_potential_V = 46500.0
sensor_plane_potential_V = 0.0
reference = "docs/elm.tex; isotropic axial EL, absorbing walls, meridional isotropic-in-core
↪  capture"

[pure_xe]
d_T_mm_sqrtbar_per_sqrtcm = 3.5
d_L_mm_sqrtbar_per_sqrtcm = 0.9
yield_gain_photons_per_electron_kV = 137.0
yield_threshold_kV_cm_bar = 0.73

[xe_he15]
d_T_mm_sqrtbar_per_sqrtcm = 1.6
d_L_mm_sqrtbar_per_sqrtcm = 0.75
yield_gain_photons_per_electron_kV = 139.0
yield_threshold_kV_cm_bar = 0.80


schema = "nausicaa.sipm_spec.v1"
response_model = "statistical"
```

```
statistical_substeps = 128
sparse_trigger_limit = 16
microcell_count = 39984
illuminated_microcell_count = 21808
microcell_pitch_m = 15e-6
recovery_s = 1.25e-08
arrival_model = "uniform_in_bucket"
illumination = "uniform_disk_subset"
recovery_pde = "constant"
initial_state = "stationary_dark"
charge_assignment = "avalanche_bucket"
input_photon_band = "blue_tpb"
shape = "square"
side_m = 0.003
pde = 0.32
gain = 360000.0
dark_reference_rate_Hz = 700000.0
dark_reference_area_m2 = 9e-6
crosstalk = false
afterpulsing = false
gain_spread = false
seed = 2026

[acquisition]
channels = "illuminated_in_event"
additional_channels = []
quiet_before_buckets = 4
quiet_after_buckets = 4
empty_first_bucket = 0
empty_last_bucket = 7

[provenance]
device = "Hamamatsu S14160-3015PS, 3x3 mm2; nominal +4 V, 25 C"
pde = "0.32 peak PDE at 460 nm used as provisional TPB-spectrum average; packaged PDE includes
↪  entrance window"
gain = "3.6e5 nominal electrons per primary avalanche; charge per fully recovered avalanche"
dark_rate = "S14160-3015PS nominal primary dark rate 700 kcps at 25 C and +4 V"
window = "S14160-3015PS silicone resin, nominal n=1.57; unit external coupling used in pilot"
datasheet = "papers/SiPM/Hamamatsu_S14160_small_cells.pdf"

recovery = "Provisional 12p5 ns exponential recovery; sensitivity parameter, not a measured 3015PS
↪  recharge constant"
illumination = "Uniform centered 2.5 mm disk on 3x3 mm square: round(39984*pi*1.25^2/9)=21808 lit
↪  cells; remaining 18176 dark-only; full 700 kcps dark rate split by cell count"

baseline_pe = 0.0
bucket_width_s = 2.5e-7
current_limit_A = 0.001
lower_clipping = false
noise = "independent_gaussian_pe"
noise_rms_pe = 3.5416666666666665
schema = "nausicaa.sipm_electronics_spec.v1"
seed = 2027
transfer = "bucket_average_current"

[provenance]
noise = "Same input charge RMS as 0.5 PE at gain 2.55e6; rescaled to gain 3.6e5"
range = "User design input: upper clipping at 1 mA; supersedes the provisional 300 PE limit"
timing = "User 0.25 us buckets; integrated charge divided by bucket duration"

schema = "nausicaa.ion_csa_spec.v1"
model_version = "v3_uniform_stationary_noise_v1"
design_card = "config/csa_design_v3.toml"
calibration_card = "config/csa_v3_response_calibration.toml"
calibration_directory = "results/ltspice/csa_v3_response_calibration"
variant = "length_1200um"
```

```
nominal_discharge_time_s = 20.0
averaging_s = 0.005
input_timing = "uniform_mean_arrivals_with_finite_gap_current"
pad_side_m = 0.0005
gap_m = 0.005
gap_speed_m_s = 3.0
noise_low_Hz = 0.0001
noise_high_Hz = 1000000.0
noise_model = "stationary_gaussian_from_one_sided_spice_output_psd"
channel_noise = "independent_pads"
noise_seed = 20261030
covariance_tail_fraction_tolerance = 0.00001
covariance_max_frequency_Hz = 10000.0
quiet_before_frames = 2
recovery_tail_s = 3.0
current_step_s = 0.000005
solver_step_s = 0.00025
simulation_prehistory_s = 0.01
initial_spice_guard_V = 0.63
control_voltage_edges_V = [0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.61, 0.62, 0.63, 0.64, 0.65, 0.675,
↪  0.7, 0.725, 0.75, 0.775, 0.8, 0.9]
controls_per_stratum = 30
linear_error_tolerance_ENC = 0.5
compression_fractions = [0.1, 0.25, 0.5, 0.75]
compression_minimum_deficit_ENC = 1.0
storage_float = "Float32_validated_against_Float64"

schema = "nausicaa.ion_csa_reconstruction.v1"
method = "three_frame_inverse"
input = "saved_mean_and_noise_realisation"
boundary_signal = "zero_collected_ions_beyond_saved_quiet_frames"
noise = "exact_covariance_transform"
nonnegative_constraint = false
signal_to_noise_cut = false
compression_fractions = [0.1, 0.25, 0.5, 0.75]
compression_minimum_deficit_ENC = 1.0
numerical_tolerance_ENC = 0.001
covariance_relative_tolerance = 0.0000000001
sampling_tolerance = 0.01

chamber_length_cm = 150.0
fano_factor = 0.15
max_events_per_sample = 8000
min_ion_drift_cm = 5.0
max_ion_drift_cm = 145.0
primary_reference = "config/specs/generated/df1414afcdc2/realisation.toml"
schema = "nausicaa.realisation_spec.v1"
seed = 2026
w_eV = 21.9

[samples]
bb0nu = "events/0nubb_hitdf.h5"
single_electrons = "events/electron_0nubb_energy_hitdf.h5"
```